EXP.1

28/7/26

STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

AIM:

To study, perform and execute various network commands that are used in linux and windows

NETWORKING COMMANDS:

1. arp -a (Linux)

ARP is short form of address resolution protocol, It will show the IP address of your computer along with the IP address and MAC address of your router

>>> _gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3

2. hostname (Linux)

It displays the name of the compiler. Simplest of all TCP/IP commands

>>> fedora

3. ipconfig /all (Windows)

displays detailed configuration information abt your TCP/IP connection including Router, Gateway, DNS DHCP and type of ethernet adapter in your system.

>>> Physical address : 0A-00-27-00.00-04
DHCP enabled : No
Autoconfiguration Enabled : Yes
IPv4 Address : 192.168.56.1 (Preferred)

4. nbtstat -a (windows)

helps solve problems with NETBIOS name resolution.

>>> NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n]
[-r] [-R] [-s] [-S] [interval]]

-a (adapter status) List the remote machine's name table given its name

-A (Adapter status) . . .

-c (cache) . . .

:

RemoteName em Rote host machine name

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5. nbstat - c (Windows)

List NBT's cache of remote [machine] names and their IP address

>>> Ethernet 2:

Node IPaddress: [192.168.56.1] Scope Id: []

No names in cache

Ethernet:

Node IPaddress: [172.16.10.183] Scope Id: []

No names in cache

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6. netstat (windows)

Displays a variety of statistics about a computers active TCP/IP connections (Including routing tables, interface statistics, network connections both incoming and outgoing, etc)

>>> Active connections

Proto	Local Address	Foreign Address	State
TCP	172.16.10.183:7680	172.16.8.179:55259	TIME_SWAIT
TCP	172.16.10.183:7680	DESKTOP-C9MTBKB: 55811	ESTABLISHED

:

7. nslookup (name server lookup) (Linux)

To perform DNS lookups in Linux can operate in two modes - in interactive and non interactive (displays DNS details - IP address of particular computer, MX records for a domain or the ns servers of a domain)

>>> nslookup www.pinterest.com

>>> Server : 127.0.0.53

Address : 127.0.0.53#53

Non-authoritative answer

:

Name : e6449.a.akamaiedge.net

Address : 23.41.121.22

8. pathping (unique to windows)

Traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics along each hop (combination of Ping and Tracert commands) on the rate of data loss

>>> Usage : pathping [-g host-list] [-h maximum_hops] [-i address] [-n] . . . [-4] [-6] target_name

options :

-g host-list Loose source route along host-list

-p period Wait period milliseconds btwn pings

:

9. ping (Packet Internet Groper) (Windows)

Best way to connectivity btwn two nodes. It uses ICMP to communicate to other devices (Internet control message protocol) test

>>> ping www.kaggle.com

>>> Pinging www.kaggle.com [35.244.233.98] with 32 bytes of data

:

Ping statistics for 35.244.233.98:

Packets: Sent = 4, Recieved = 4, Lost = 0 (0% Loss)

Approximate Round trip times in milli-seconds:

Minimum = 3ms, Maximum = 4ms, Average = 3ms

>>> ping localhost #syntax: ping hostname

>>> Pinging HDC04222258 [::1] with 32 bytes of data

:

Ping statistics for ::1:

Packets: Sent = 4, Recieved = 4, Lost = 0 (0% loss)

Approximate round trips in milliseconds:

Minimum = 0 ms, Maximum oms, Average = oms

>>> ping 172.16.10.183 #ping IP address

>>> Pinging 172.16.10.183 with 32 bytes of data

:

Approximate round trips times in milli-seconds:

Minimum = 0 ms, Maximum = 0 ms, Average = 0 ms

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10. Route: (windows)

To show/manipulate the IP routing table.

Primarily used to setup static routes or specific host or networks via an interface

>>> ROUTE [-f] [-P] [-4|-6] command [destination]

-f Clears the routing table of all gateway entries . . . . .

Important Linux Commands

1. IP

Basic command used by administrator. It shows address information, manipulate routing, plus display network various devices, interfaces and tunnels

SYNTAX - ip <OPTIONS> <OBJECT> <COMMAND>

>>> ip address show

>>> 1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noqueue state unknown group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

>>> ip route get 10.10.1.4

>>> 10.10.1.4 via 10.2.2 dev enp0s3 src 10.0.2.15 uid 1000 cache

2. ifconfig

It is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by IP command

>>> enp0s3: flags = 4163 <UP, BROADCAST, RUNNING, MULTICAST> mtu 1500

inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255

3. mtr (Matt's traceroute)

It is a program with a command-line interface that serves as a network diagnostics tool and troubleshooting tool combines the functionality of ping and traceback commands.

Syntax : mtr <options> hostname/IP

>>> mtr google.com Shows the statistics

>>> fedora (10.0.2.15) → google.com (142.251.221.206)

Keys : Help Display mode Restart Statistics Order of fields quit

Host	Packets Loss%	Snt	Pings Last	Avg	Best
1. pnmaa-ba-in-f14.1e100.net	0.0%	89	7.1	7.2	5.6

>>> mtr -g google.com - Shows numeric IP addresses and hostnames

>>> fedora (10.0.2.15) → google.com (142.251.221.206)

Keys : Help Display mode Restart statistics Order of files quit

Host	Packets Loss%	Snt	Ping Last	Avg	Bst
1. pnmaa-ba-in-f14.1e100.net	0.0%	162	16.1	8.2	6.3

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4. tcpdump

Designed for capturing and displaying packets

To install tcpdump → dnf install -y tcpdump

>>> tcpdump -D

>>> 1. enp0s3 [UP, Running, connected]

2. any (Pseudo device that captures on all interfaces)

3. [UP, Running]

⋮

>>> tcpdump -i eth0

↳ To capture traffic on eth0

5. ping

It is a tool that verifies IP level connectivity to another TCP/IP computer by sending Internet control Message Protocol (ICMP) Echo request messages. It is the primary TCP/IP command used to troubleshoot connectivity, reachability & name resolution

```
>>> ping google.com
>>> PING google.com (142.250.207.78) 56(84) bytes of data
    64 bytes from hkg12s32-in-f14.1e100.net (142.250.207.78):
    :                                    icmp_seq=1 ttl=255
                                         time=627 ms
```

Stops by ctrl+c

```
>>> ping -c 10 google.com          → pings a host 10 times
>>> PING google.com (142.251.221.206) 56(84) bytes of data
    64 bytes from pnmaa-ba-in-f14.1e100.net (142.251.221.206):
    :                                    icmp_seq=1 ttl=256 time=6.91 ms
    :
    :
    :
```

Configuring an Ethernet connection by using nmcli

If you connect a host to the network over Ethernet, you can manage the connection's settings on the command line by using the nmcli utility

Procedure

1. List the network manager connection profiles

   #nmcli connection show

NAME	UUID	TYPE	DEVICE
wired connection 1	a5e b6490-cc20-3668-81f8	Ethernet	enp

2. #nmcli connection add con-name <connection-name> ifname <device-name> type ethernet

3. Rename the connection profile (optional)

   #nmcli connection modify "wired connection 1"

   Here wired connection 1 is the name of the connection

4. Display the current settings of the connection profile

   # nmcli connection show

5. Configure the IPv4 settings:

   To use DHCP, enter:

   #nmcli connection modify "wired Connection1" ipv4.method auto

   To set a static IPv4 address, network mask, default gateway, DNS servers and search domain enter

#nmcli connection modify "wired Connection 1" ipv4.method manual ipv4.addresses 192.02.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com

6. Configure the IPv6 settings

To use stateless address autoconfiguration (SLAAC), enter:

#nmcli connection modify wired connection 1

ipv6.method auto

To set a static IPv6 address, network mask default gateway, DNS servers and search domain, enter:

#nmcli connection modify 'wired connection 1' ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::ffe ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com

7. Activate the profile

#nmcli connection up Internal-Lan

Verification

1. Display the IP settings of the NIC

#ip address show enp1s0

2. Display the ipv4 default gateway: #ip route show default

default via 192.0.2.254 dev enp1s0 proto static metric 102

3. Display the IPv6 default gateway: #ip-6 route show default

default via 2001:db8:1::ffee dev enp1s0 proto static metric 102 pref medium

4. Dyplay the DNS

Settings : #cat

/etc/resolv.conf
search example.com
nameserver 192.0.2.200
nameserver 2001:db8:1::ffbb

If multiple connection profiles are active at the same time, the order of nameserver entries depend on the DNS priority values in the profile and the connection types

5. Use the ping utility to verify that this host can send packets to other

hosts : # ping <host-name-or-IP address>

Troubleshooting :

- Verify that the network cable is plugged-in to the host and a switch
- check whether the link failure exists only on this host or also on other hosts connected to the same switch
- Verify that the network cable and the network interface are working as expected. Perform hardware diagnosis steps and replace defect cables and network interface cards.
- If the configuration on the disk does not match the configuration on the device, starting or restarting Network Manager creates an in-memory connection that reflects the configuration of the device.

Student Observation :

1. Which command is used to find the reachability of a host machine from your device?

Ping command used to check if a remote host is reachable from your computer. It sends ICMP echo request packets to destiny and wait for reply

Command: ping google.com

2. Which command gives the details of hops taken by a packet to reach its distination?

The Traceroute command displays the path (hops) that packets take from the source system to the distination. It also shows the time taken at each hop.

command: traceroute google.com

3. Which command displays the IP configuration of your machine?

ifconfig displays the network interface config

command: ifconfig or ipaddr show.

4. Which command displays the TCP ports sts in your machine?

The netstat command display active TCP connection, listening ports & networks

command: netstat -t, ss -t, netstat -tln, ss-tln

5. Write command to modify the IP configuration in a Linux machine?

The ifconfig or IP is used to assign or modify the IP configuration in a Linux machine

command: Sudo ip addr replace 192.168.1.100/2

dev. enp0s3